

Manual

HYDRA Interfacing Module for SAP PP Serial Production SAP-PPREM 8.2

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1 HYDRA Interfacing Module for SAP PP Serial Production

Summary

Fields of application

If serial production is used in SAP R/3 the HKMPP-REM interface allows HYRA to be used as lower-level system to collect actual data during the production process.

It is typical for serial production that production plans are created and processed in relation to time and quantities. Compared to the individual batch and order-related production control, the efforts required for controlling serial production are reduced considerably and actual data recording is simplified.

The costs accruing with serial production are booked on a product cost collector. As a part of actual data collection, the product cost collector is, for example, charged with the costs for material and production. Consequently, a goods receipt posting discharges the product cost collector. The product cost collector is created for a material in a specific production version within a plant.

Only the produced quantities are uploaded in serial production. The withdrawn components and rendered services are normally posted automatically.

In HYDRA actual data collection is not based on a fixed pattern. In fact data can be collected at any time. In addition to this, uploads to SAP may also be configured individually, depending on how up-to-date data are expected to be. This ensures a simple integration into existing production and collection processes.

Normally, component consumption is calculated automatically from the BOM and withdrawn from stock in a retrograde manner. Production times are taken from the work plan/routing. While posting, material and production costs are updated on the product cost collector.

In case of longer processing times, actual data can also be recorded at counting points within the production line to be able to post consumptions more closely to the physical goods receipt posting. To do so, it has to be defined for the serial production profile that a counting point posting may be performed for the material. Furthermore, the control key has to identify an operation as the counting point (milestone) in the work plan.

SAP R/3 provides BAPIs for the upload of recorded data of serial orders from third-party systems. They have virtually been designed as supplement for SAP-GUI. Using these BAPIs, actual data can be recorded in HYDRA, transferred to SAP where they are processed, as if they were directly entered in SAP-GUI. But HYDRA has to provide for a corresponding data basis, which can be achieved by downloading the relevant data.

Implementation notes

You use the function package SAP-PPREM if you wish

- to use serial production in SAP ECC and do not want to use production orders
- to upload quantities relating to the production version from MES to SAP
- to use production order handling in SAP ECC but also want to plan or show planned orders, as this provides a larger period of forward planning

Integration

If you use the SAP-PPREM component the orders/operations transferred by it represent the posting framework for many other HYDRA components.

Features

- Transfer of planned orders using the IDoc type LOIPLO01
- Comparison of new planned orders with the planned orders that already exist in HYDRA by date/workplace/header material
- Collection of actual quantities (yield and scrap) in HYDRA
- Upload of actual quantities to SAP PP-REM using the SAP BAPI RepManConfirmation1.CreateMTS as counting point upload
- Notification if an upload could not be posted (prerequisite: SIS-ESK).

2 Mapping of SAP-PPREM in HYDRA

Usage

In the context of connecting HYDRA to SAP PM-REM, it is the task of HYDRA to collect actual data in the production process and to upload the collected actual quantities (yield and scrap quantities) to SAP PP-REM. Requirements represent the data basis in SAP R/3. With regard to these requirements, planned orders are created as replenishment elements in the course of the MRP run in SAP. These created planned orders are adopted from SAP and are used as the target for production in relation to time, date and the quantity to be produced.

The download of planned orders is initiated cyclically by R/3. The data are stored in an IDoc (intermediate document) and uploaded to HYDRA. In general, planned orders are not a constant unit. With each planning run, requirements are re-calculated and new planned orders are created as replenishment elements. Planned orders are transferred discretionally to HYDRA, i.e. for a transferred planned order number, the system creates an order or operation record in HYDRA or runs an update for these values if such records exist already. After order data adoption by HYDRA, the planned orders are ready for collection as production orders of the order type "REM".

Uploading the confirmations from HYDRA is controlled in accordance with user requirements. In the course of the confirmation, yield and scrap quantities are primarily transferred to SAP. The transfer of recorded actual times is not supported by SAP.

Communication with SAP R/3 takes place via two technical interfaces:

Download of planned orders via LO-SCI:

IDoc type:	LOIPLO01
Segment Type:	E1PLAFL
	E1PLOPL
	E1PLUVL
	E1KBEDL
	E1RESBL

Upload of confirmations for storage scenario:

BAPI:	RepManConfirmation1
Method:	CreateMTS

Planned Orders Download SAP → HYDRA

The download of data to HYDRA is initiated by R/3 through SAP workflow processes. The download of planned orders is performed via the LO-SCI interface by means of IDoc LOIPLO01. The transfer is always fully completed, which means that all planned orders available for the selection period chosen in SAP are transferred to HYDRA. In HYDRA, these planned orders are created with the order type "REM".

Confirmation Upload HYDRA → SAP

The basis for confirmation to SAP are the log records collected in HYDRA. Depending on the configuration, these may be T/U/E or H records. Collected yield and scrap quantities are uploaded cyclically in accordance with the configuration in the HYDRA Scheduler. Confirmation is made through the BAPI RepManConfirmation using the CreateMTS method via synchronous RFC (sRFC).

3 Data type definitions

Type	Description
CHAR x	Information is left-aligned for the data type CHAR. Places that are not required are filled with blanks (U+0020). If the field is not used, it must be completely prepopulated with blanks. Example: "ABCD "
NUM x	Numeric field of the length x without sign. The data type NUMC only supports digits (ASCII characters 30 Hex to 39 Hex). These digits are right-aligned and unnecessary places are filled with zeros. If the field is not used, it must be completely prepopulated with zeros (U+0030). Example: "00000002"
DEC x.y QUAN x.y	Numeric field of the length x and y decimal places. A data field in HYDRA format is preceded by a sign ("+" or "-") and includes a decimal point. Places that are not required are filled with zeros. If the field is not used, it must be completely prepopulated with zeros (U+0030) including algebraic sign and decimal separator. e.g. DEC 13,3: -1234567890,123 → -1234567890.123 234567890,3 → +0234567890.300 Note: The field length is indicated WITHOUT algebraic sign and WITHOUT decimal point in the tabular description of the structure. This means, for example, that a field QUAN 13.3 is converted to an external length of CHAR15.
DATE	Format YYYYMMDD. If the field is not used, it must remain empty (filled with blanks (U+0020)).
TIME	Format HHMMSS. If the field is not used, it must be set to "000000" (zeros with (U+0030)).



HYDRA does not support special characters for all alphanumeric fields. This includes, among others: "\" (backslash), "|" (pipe), „“““ (double quotes), and " ’ “ (single quotes). You cannot enter these characters using shop floor terminals and the MOC does not support them.



The character " ; " (semicolon) is used as a separator in the system. You must not use this character in key fields (e.g. order/operation number, MES batch number, personnel number).



The character " % " (percent) is used as a placeholder for database communication. You should not use this character to prevent the result from being falsified.

4 Downloading Operation Data

Overview

Message type: LOIPLO
 IDoc type: LOIPLO01
 Segments: E2PLAFL001
 E2PLOPL001
 E2KBEDL
 E2RESBL

Segment E2PLAFL001

Segment E2PLAFL001 includes data of the planned order header.

Field name	T	L	D	Description	Usage in HYDRA
ABMNG	QUAN	14	3	Reduced quantity in planned order	Not used
AUFFX ANR.FU:29	CHAR	1		Fixing flag in planned order data	Not used
AUFNR	CHAR	12		Serial order number	Not used
AVMNG ANR.SGR:AUSB	QUAN	14		Planned scrap quantity	OH.Target scrap in base quantity unit
BESKZ	CHAR	1		Procurement type	Not used
DISPO ANR.DISP	CHAR	3		Material controller	OH.MRP controller
EKORG	CHAR	4		Purchasing organization	Not used
GSBTR	DATS	8		Total confirmation date of planned order after ATP check of components	Not used
GLTRS	DATS	8		Scheduled end time	Not used
GLUZS	TIMS	6		Earliest scheduled end time: Execution	Not used
GSMNG ANR.SGR:GUTB	QUAN	14		Planned order quantity	OH.Target quantity in base quantity unit
GSTRS	DATS	8		Scheduled start time	Not used
GSUZS	TIMS	6		Earliest scheduled start time: Execution (time)	Not used
KAPFX	CHAR	1		Flag: Planned order - capacities planned	Not used
KDAUF ANR.KDAUF	CHAR	10		Sales order number	OH.Sales order
KDPOS ANR.KDPOS	CHAR	6		Item number in sales order	OH.Sales order item
LGORT ANR.LGORT	CHAR	4		Storage location	To be used as receiving storage location in confirmation
MATNR ANR.ATK	CHAR	18		Material number	OH.Article
MDACC	CHAR	4		Planned order handling action	Not used

Field name	T	L	D	Description	Usage in HYDRA
MDACD	DATS	8		Planned order action date	Not used
MDACH	CHAR	2		Planned order handling action control	Not used
MDACT	TIMS	6		Planned order handling action time	Not used
MDPBV	CHAR	1		Planned order confirmation availability	Not used
MEINS ANR.SGE:GUTB	UNIT	3		Base quantity unit	OH.Base quantity unit OP.Base quantity unit
PAART ANR.FU:45	CHAR	4		Order type	Not used
PALTR	DATS	8		Resolution date (explosion date)	Not used
PEDTR ANR.DATSE	DATS	8		Order finish date in planned order	OH.Order finish date
PERTR	DATS	8		Planned creation date in planned order	Not used
PLNAL	CHAR	2		Group counter	OH.Work plan
PLNNR ANR.APNR	CHAR	8		Key of planned group	OH.Work plan
PLNTY	CHAR	1		Planned type	Not used
PLNUM ANR.SAPAUNR	CHAR	10		Planned order number	HYDRA order number according to configuration (*1)
PLSCN	NUMC	3		Long-term planning scenario	Not used
PLWRK	CHAR	4		Planning plant	Not used
PSPEL ANR:PRJNR	NUMC	24		PSP element	OH.Project number
PSTTR ANR.DATFB	DATS	8		Order start date in planned order	OH.Order start date
PWWRK ANR:WERK:S	CHAR	4		Production plant in planned order	Used as actual plant in confirmation
SEQNR	NUMC	14		Order sequence number	Not used
SERNR	CHAR	8		Serial number	Not used
SOBES	CHAR	1		Special procurement type	Not used
SOBKZ	CHAR	1		Special inventory flag	Not used
STLFX	CHAR	1		Fixing flag for BOM explosion	Not used
UMSKZ	CHAR	1		Implementation flag of planned order	Not used
VERID ANR.FERTVER	CHAR	4		Production version	Stored for confirmation
VFMNG	QUAN	14		Planned order of confirmed quantity	Not used
WEBAZ	DEC	5		Processing time for goods receipt in days	Not used
MATNR_EXTERNAL	CHAR	40		Long material number	Not used
MATNR_VERSION	CHAR	10		Version number for MATNR field	Not used
MATNR_GUID	CHAR	32		External GUID for MATNR field	Not used

Segment E2PLOPL001

The segment E2PLOPL001 includes operation data.

Field name	T	L	D	Description	Usage in HYDRA
VORN ANR.SAPVGNR	CHAR	4		Operation number	HYDRA operation number according to configuration (*1)
PVZNR	CHAR	4		Superordinate operation	Not used
ARBEH	UNIT	3		Work unit	Not used
ARBEI	QUAN	8		Operation work	Not used
ARBID ANR.MNR	NUMC	8		Workplace ID	Target workplace OP.Machine and/or OP.Machine group (see note)
DAUNE	UNIT	3		Unit duration normal	Not used
DAUNO	QUAN	6		Normal operation duration	Not used
FSEDD	DATS	8		Earliest scheduled end date: Execution (date)	Not used
FSEDZ	TIMS	6		Earliest scheduled end time: Execution (time)	Not used
FSAVD	DATS	8		Earliest scheduled start date: Execution (date)	Not used
FSAVZ	TIMS	6		Earliest scheduled start time: Execution (time)	Not used
FSSAD	DATS	8		Earliest scheduled start date: tear down/retool (date)	Not used
FSSAZ	TIMS	6		Earliest scheduled start time: tear down/retool (time)	Not used
FSSBD	DATS	8		Earliest scheduled start date: Processing (date)	Not used
FSSBZ	TIMS	6		Earliest scheduled start time: Processing (time)	Not used
PLNFL	CHAR	6		Sequence	Not used
SSEDD	DATS	8		Latest scheduled end date: Execution (date)	Not used
SSEDZ	TIMS	6		Latest scheduled end time: Execution (time)	Not used
SSAVD	DATS	8		Latest scheduled start date: Execution (date)	Not used
SSVAZ	TIMS	6		Latest scheduled start time: Execution (time)	Not used
SSSAD	DATS	8		Latest scheduled start date: tear down/retool (date)	Not used
SSSAZ	TIMS	6		Latest scheduled start time: tear down/retool (time)	Not used
SSSBD	DATS	8		Latest scheduled start date: Processing (date)	Not used
SSSBZ	TIMS	6		Latest scheduled start time: Processing (time)	Not used
USR04 ANR.TLG	QUAN	15		User field 04	Partitioning (see note)
USR05 ANR.SZY	QUAN	15		User field 05	Target cycle (see note)
USE04	UNIT	3		Unit user field 04	Not used
USE05	UNIT	3		Unit user field 04	Not used
VGE01	UNIT	3		Unit of activity 1	Not used
VGE02	UNIT	3		Unit of activity 2	Not used
VGE03	UNIT	3		Unit of activity 3	Not used
VGE04	UNIT	3		Unit of activity 4	Not used
VGE05	UNIT	3		Unit of activity 5	Not used

Field name	T	L	D	Description	Usage in HYDRA
VE06	UNIT	3		Unit of activity 6	Not used
VGW01 ANR.VGW01	QUAN	10		Default value activity 1	Not used
VGW02 ANR.VGW02	QUAN	10		Default value activity 2	Not used
VGW03 ANR.VGW03	QUAN	10		Default value activity 3	Not used
VGW04 ANR.VGW04	QUAN	10		Default value activity 4	Not used
VGW05 ANR.VGW05	QUAN	10		Default value activity 5	Not used
VGW06 ANR.VGW06	QUAN	10		Default value activity 6	Not used
XDISP	CHAR	1		Flag: Operation/KBED is planned	Not used
ANZMA ANR.MBVERH:NORM	DEC	7	2	Number of employees	Not used

Segment E2KBEDL

Field name	T	L	D	Description	Usage in HYDRA
BEDID	NUMC	12		Serial number of capacity requirements record	Not used
BEDZL	NUMC	8		Internal counter	Not used
CANUM	NUMC	4		Counter of capacity requirements record	Not used
BEDKZ	CHAR	1		Availability indicator	Not used
KABRREST	FTP	22		Remaining quantity clearing/retooling capacity demand	Not used
KABRSOLL	FTP	22		Clearing/retooling capacity demand	Not used
KAPAR	CHAR	3		Capacity category	Not used
KAPID	NUMC	8		Capacity (number)	Only records of capacity type "001" are transferred.
KBEAREST	FTP	22		Remaining quantity processing capacity requirements	Not used
KBEASOLL	FTP	22		Processing capacity requirements	Processing duration (target for RPA11)
KEINH	UNIT	3		Unit of capacity requirements	Unit for processing time and setup time (*2)
KRUEREST	FTP	22		Remaining quantity setup capacity requirements	Not used
KRUESOLL	FTP	22		Setup capacity requirements	Setup duration (target for RPA7) (*2)

OH.xxx Field xxx in HYDRA order header

OP.xxx Field xxx in HYDRA operation

- (*1) For more information on generating the HYDRA order number, please see below.
- (*2) The contents of these fields must not exceed a duration of 590 hours.

Notes on the operation structure

HYDRA order number (*1)

The HYDRA order number consists of a configurable part of the SAP key fields **E2PLAFL.PLNUM**, **E2PLOPL.VORNR**. The following steps are carried out to make sure the order number is transferred to HYDRA in a standardized manner.

- When the CHAR fields are transferred, the system converts blank characters to "0", since the barcode is also assigned "0" instead of blank characters.
- Two leading places are added to the SAP order number in order to calculate the HYDRA order number. Consequently, HYDRA uses a number that is extended by two places compared to the original SAP planned order number. This is taken into account if quantities are uploaded to SAP for the planned order (series production).
- By default, the following conventions apply:
→ PLNUM (=order) C8, VORNR (=operation) C4.

Operation quantity

The SAP IDoc type LOIPLO01 does not provide an explicit operation quantity. SAP only provides a target yield and a target scrap quantity in the order header. This quantity is transferred to each operation that is sent with this order. Any scrap quantities of the preceding operations cannot be integrated.

Overview - how the user fields are used:

By default, HYDRA processes the user-specific fields USR04, USR05, USE04 and USE05 as follows:

User field 04 (USR04):

HYDRA partitioning/cavity

User field 05 (USR05):

HYDRA target cycle in seconds per 1000 machine strokes

User field 04 unit (USE04):

Not used

User field 05 unit (USE05):

Not used

HYDRA partitioning

If the user field USR04 includes a numeric value, the system enters this value as the partitioning in the operation.

If this is not the case or if the value is "0", the system automatically enters the value 1 as the partitioning.

BDE cycle time

The system attempts to take the target cycle from the field E2PLOPL001.USR05. If this is not possible as no target cycle is entered (i.e. the field is "0"), the system uses the following formula to calculate the target cycle using the processing time value of the segment E2KBEDL.KBEASOLL. To do so, this segment must include a capacity type record "001":
$$\text{int}((\text{E2KBEDL.KBEASOLL} / (\text{E2PLAFL.GSMNG} / \text{E2PLOPL001.USR04})) * 1000)$$

Standard times

In HYDRA the content of the field E2KBEDL.KBEASOLL specifies the standard time for machine assignment (target for RPA 11).

In HYDRA the content of the field "E2KBEDL.KRUESOLL" specifies the standard time for the machine setup (target for RPA 7).

Information on the workplace - ARBID field:

- The workplace ID (ARBID) is not the workplace number as it is stored in the work plan but the unique ID of the workplace. Data is collected in HYDRA and uploaded to SAP R/3 PP-REM for the workplace entered in this field.

5 Uploads

Usage

Quantities recorded in HYDRA can be uploaded automatically to SAP series production where they can be posted. These are mere quantity postings. SAP does not provide for the transfer of actual SAP activities.

Uploads are performed by synchronous RFC (sRFC). SAP posts the uploads by calling the BAPI RepManConfirmation1 using the CreateMTS method. If posting cannot take place in SAP, it will be recorded in an error log. The event is generated ([SAP.OUTBOUND_FM_POST_ERROR](#)) if HYDRA Escalation Management is used for HYDRA MES Link Enabling (license SIS-ESK and SAP-ESK).

Record types supported by HYDRA

HYDRA-BDE uploads recorded quantities (yield and scrap) to SAP R/3 PP. The customizing settings configured for the HYDRA order type determine which record types represent the basis for HYDRA uploads.

If the upload of partial quantity uploads is enabled the quantities will be taken from the generated T records. The field SCRAPREASON of the upload structure is assigned the recorded scrap reason. The U records are not uploaded in this case. The generated E record is transferred with the quantities of the generated T record, provided that quantities have been recorded at all. Yield and scrap are each uploaded in individual records to SAP R/3.

If partial quantity uploads are not configured for being uploaded, the quantities will be taken from the interrupted operations (U record) or logged off operations (E record) recorded in HYDRA. Scrap reasons are not transferred. Yield and scrap are each uploaded in individual records to SAP R/3.

Records having the quantity "0" are not uploaded to SAP, irrespective of whether the upload of partial uploads has been enabled or not. However, this can also result in the final upload of an operation not to be transferred to SAP, as this posting does not include a quantity.

Upload structures of the BAPI RepManConfirmation1

Upload structure BflushFlags

This structure specifies the type of upload.

Field	T	L	D	Description	Usage in HYDRA		
-------	---	---	---	-------------	----------------	--	--

BCKFLTYPE	CHAR	2		Backflushing type of a BAPI backflush (upload)	"02" - has to be discussed in detail with the customer!		
RP_SCRAPTYPE	CHAR	1		Scrap type for reporting point scrap backflush (upload)	Fixed "1" scrap at the specified-reporting point		
ACTIVITIES_TYPE	CHAR	1		Scope of the separated activity backflush (upload)	Not used.		
COMPONENTS_TYPE	CHAR	1		Scope of the separated goods issue posting	Not used.		

Upload structure BFlushDataMTS

The structure includes the reporting point for which the data included in the structure BflushdataGen apply.

Field	T	L	D	Description	Usage in HYDRA		
REPPPOINT	CHAR	4		Reporting point	Operation number		

Upload structure BFlushDataGen

The structure BflushdataGen transfers actual user data to SAP.

Field	T	L	D	Description	Usage in HYDRA		
PDC_NUMBER	CHAR	12		PDC number (unique ID for all PDC systems)	"HY"+reference from ADE_PROTOKOLL		
MATERIALNR ANR.ATK	CHAR	18		Material number	Material number from planned order Ak.Artikelnummer		
PRODPLANT ANR.WERK:S	CHAR	4		Plant	Specified plant		
PLANPLANT ANR.WERK:S	CHAR	4		Planning plant	Specified plant		
STORAGELOC ANR.LGORT	CHAR	4		Receiving storage location for repetitive manufacturing	Specified storage location		
PRODVERSION ANR.FERTVER	CHAR	4		Production version	Production version from planned order		
PRODLINE	CHAR	8		Production line for repetitive manufacturing	Not used.		
PLANNINGID	CHAR	8		Planning ID 2	Not used.		
BATCH	CHAR	10		Receiving batch for repetitive manufacturing	Not used.		
POSTDATE ADEPRO.SKDAT	DATS	8		Posting date in the document	Shift date of the posting		
DOCDATE	DATS	8		Document date in document	Upload date		
DOCHEADERTXT	CHAR	25		Document header text	Not used.		
BACKFLQUANT ADEPRO.EGR:GUTP	QUAN	13	3	Quantity in unit of entry	Yield in primary quantity unit		
SCRAPQUANT ADEPRO.EGR:AUSP	QUAN	13	3	Scrap quantity	Scrap quantity in primary quantity unit		

Field	T	L	D	Description	Usage in HYDRA		
UNITOFMEASURE ADEPRO.EGE:GUTP	UNIT	3		Unit of measure for backflush quantity and for actual scrap	OP target quantity unit of the primary quantity		
UNITOFMEASURE_ISO	CHAR	3		ISO code for unit of measurement	Not used.		
SCRAPREASON	CHAR	4		Reason for scrap	Scrap reason (when transferring partial uploads)		
REVLEVEL	CHAR	2		Revision level	Not used.		
PLANORDER	CHAR	10		Planned order number	SAP planned order according to specifications		
ORDERCOSTS	CHAR	1		Indicator: Post with order costs (lot-size independent)	Not used.		
INCLCOMPSCRAP	CHAR	1		Indicator: Post with component scrap	Not used.		
MATERIALNR_EXTERNAL	CHAR	40		Long material number (future development) for the field MATER	Not used.		
MATERIALNR_GUID	CHAR	32		External GUID (future development) for the field MATERIALNR	Not used.		
MATERIALNR_VERSION	CHAR	10		Version number (future development) for the field MATERIALNR	Not used.		

Please note:

It is not possible to upload quantities using partial uploads and to record them by the total quantity counter at MDE machines at the same time. This input type can result in negative quantity postings for the yield if OPs are finished.

This restriction does no longer apply, if it is possible to process such negative postings (e.g. by using the SAP standard BAPI or customized processing).

6 Application-Relevant Settings in SAP

Define production scheduler

Scheduling settings for planned orders are defined for each plant/order type and production scheduler. The production scheduler can be edited as a part of the customizing in *SPRO* → *Production* → *Production control* → *master data* → *routing data* → *define production scheduler* (OPJ8).

Configure scheduling parameters

Scheduling options can now be configured for planned orders by the production scheduler. This is also performed in customizing using *SPRO* → *Production* → *capacity planning* → *master data* → *operations* → *scheduling* → *define scheduling parameters for planned orders* (OPU5).

If the scheduling level “by detailed scheduling” is configured there operations will also be transferred when downloading planned orders.

Configure serial production profile

The serial production profile defines how uploads are to be performed (counting point upload/post services...). This is configured in customizing using *SPRO* → *Production* → *serial production* → *control* → *define serial production profiles*.

Definition of user fields

Selected user fields of the work plan/operation are transferred from SAP by the PP-REM interface. HYDRA provides a default interpretation for these fields. A user field key has to be defined and saved for the operation within the work plan to be able to define values in user fields in SAP.

User field keys are defined in customizing using *SPRO* → *production* → *production control* → *master data* → *work plan data* *define user fields* (OPEC).

A meaning is to be defined for the following user fields:

SAP user field	SAP user field in the download structure	Meaning
USR04	USR04	Partitioning
USR01	USR05	Target cycle

The created user field key and corresponding values are to be defined for the operation within the routing.

Maintenance of the SAP partner agreement/profile – outbound

Maintain the following settings for outbound processing of the SAP partner agreement/profile (WE20):

Parameter name	Value
To download production orders	
Partner number	Created logical system
Partner type	LS
Message type	LOIPLO
Recipient port	Created port
Package size	1
Output mode	Transfer IDoc immediately
Basis type	LOIPLO01

Maintenance of the SAP distribution model – outbound

Parameter name	Value
To download planned orders	
Model view	Created model view
Sender / client	Logical system of the client
Recipient/erver	Logical system for the recipient system
Message type	LOIPLO

Hide unnecessary segments

If specific segments of the IDoc are not to be transferred (e.g. the segment for the components is not to be transferred as they are only transferred in relation to the header) this can be realized as a part of the customizing using the transaction BD56.

However, the SAP partner agreement/profile has to be maintained to be able to use this configuration.

IDoc enhancement

SAP standard provides the option to enhance the IDoc by customer-specific data without modification:

Enhancement	LOI00001
User exit	EXIT_SAPLLOI1_001

Planning of relevant jobs

The following programs/ reports must be planned as job to ensure that the PP-REM interface will operate automatically:

Program / report	Meaning	Note
RCCLTRAN	Start of the download for planned orders according to the selection	Planning as variant

Relevant transactions

Transaction	Meaning	Note
POIT	Start of the download for planned orders	-

7 Application-Relevant Settings in HYDRA

Maintenance of the HYDRA distribution model – inbound

Maintain entries for HYDRA inbound processing in the [HYDRA distribution model](#):

Parameter name	Value
To process production orders	
Message type	LOIPLO
Priority	None
Command	mle72imp.scr
Command parameter	/VARIANTE=< MLE variant to be used >
Description	PP-REM– Download of planned orders
Log. Target system	Created logical system
Retention period	10

Configuration of segment sorting

It might be required to re-sort segments if the SAP message type LOIPLO is enhanced using the user exit that is provided by default in SAP.

The SAP user exit allows to transfer additional data (= additional segments). However, this is only possible directly below the order header segment. If, for example, a component record is transferred at this position it cannot be posted, as the operations created by the standard segment are not yet available.

A configuration in the HYDRA INI configuration file allows to enhance the segment number (on which sequential processing is based) by a prefix before this number is inserted in the database and, as a result, to enable an alternative sorting.

The transfer of the planned order number is enabled as follows in the [HYDRA-INI configuration](#):

Parameter name	Value
INI name	HYALESRV

Parameter name	Value
Section	<MESTYP>_SORT e.g. LOIPLO_SORT
Key	<Segment name> e.g. Z2BAPI000
Value	<prefix> e.g. Z
Active	Yes
Comment	PP-REM – setting of the segment sorting



The service “HYDRA<client number> MLE-Server SAP 1” has to be restarted to activate the configuration.

Maintenance of the HYDRA distribution model – outbound

Edit an entry for HYDRA outbound processing in the [HYDRA distribution model](#):

Parameter name	Value
To upload time tickets	
Message type	REPMANCONFIRMATION1_CREATEMTS
Description	PP-REM – Upload
IDoc type	BFLUSHDATAMTS
Retention period	10
Log. target system	Created logical system
Segment name 1	BFLUSHDATAMTS

Maintenance of the HYDRA Scheduler

Edit entries for HYDRA outbound processing in the [HYDRA Scheduler](#):

Parameter name	Value
For uploads from the application to the MLE outbound transactions	
Product key	SAP-PPREM
License key	SAP-PPREM
Command (Windows):	sh.exe ./.myerprck.scr /MESTYP=REPMANCONFIRMATION1_CREATEMTS /KAT=FA /RMTYP=REM
Command (Unix):	./myerprck.scr /MESTYP=REPMANCONFIRMATION1_CREATEMTS /KAT=FA /RMTYP=REM
Comment:	SAP-PPREM: upload → MLE outbound transactions
Interval	5
For uploads from MLE outbound transactions to SAP PP-REM	
Product key	SAP-PPREM
License key	SAP-PPREM
Command (Windows):	sh.exe ./hysapupl.scr /UPLSEGNAM=BFLUSHDATAMTS /SINGLE_IDOC /SUBLEVEL=2 /SUBPROT=ON
Command (Unix):	./hysapupl.scr /UPLSEGNAM=BFLUSHDATAMTS /SINGLE_IDOC /SUBLEVEL=2 /SUBPROT=ON
Comment:	SAP-PPREM: Upload MLE outbound transactions - _> SAP
Interval	5



Please proceed as follows if you use the upload to SAP PP-REM on a HYDRA system at the same time as the upload to SAP PP using PP-PDC:

Maintain the upload type “PP” (customizing) for production orders at the [HYDRA order type](#).

Add the parameter “/RMTYP=PP” for calling the upload of time tickets to SAP using PP-PDC in the [HYDRA Scheduler](#)

Keep the upload type “REM” for planned orders/serial production at the [HYDRA order type](#).

Maintain the script myerprck.scr by the parameter “/RMTYP=REM” as specified above for calling uploads to SAP serial production in the [HYDRA Scheduler](#)

Configuration of uploads

The BAPI used for uploads to SAP serial production supports several upload modes. They can be configured subject to the requirements in HYDRA:

Definition of the reference to the planned order

If the upload is to be performed in relation to the planned order this can be set by the die [HYDRA-INI configuration](#). By default, uploads are performed without indicating the planned order number.

In case the planned order is uploaded at the same time as the planned order quantity, posting errors will be the result, as the planned order might no longer exist at this point in time.

The transfer of the planned order number is enabled in the [HYDRA-INI configuration](#) as follows:

Parameter name	Value
INI name	PP-REM
Section	BFLUSHDATAGEN
Key	PLANORDER
Value	TRANSFER transfer of the planned order SUPPRESS the planned order is not transferred (by default)
Active	Yes
Comment	PP-REM: activation of the transfer of the planned order number

Define upload type

It is up to your decision whether you use a counting point upload (by default) or an end upload.

If you decide in favor of a counting point upload the operation number will be transferred with the upload.

With the counting point upload the SAP systems withdraws all material components consumed at the uploaded counting point process in a retrograde manner, posts the services accrued for the uploaded counting point operation, reduces secondary requirements of the planned orders and updates the information included in the logistic information system.

With counting point uploads scrap is uploaded as scrap for the specified counting point (BFLUSHFLAGS.RP_SCRAPTYPE = „1“).

The end upload function is used if you want to perform the upload and actual data collection at the end of the production process.

The upload type is defined as described-below in the [HYDRA-INI configuration](#):

Parameter name	Value
INI name	PP-REM
Section	BCKFLTYPE
Key	BCKFLTYPE
Value	01 upload as end upload without reference to the operation 02 upload as counting point upload with reference to the operation
Active	Yes
Comment	PP-REM: definition of the upload type

8 Escalation SAP.OUTBOUND_FM_POST_ERROR

Usage

If an error occurs while uploading data for serial production, the application will provide an event [SAP.OUTBOUND_FM_POST_ERROR](#) that can be evaluated and forwarded by Escalation Management.

As part of serial production, the escalation fields are assigned as follows:

Prerequisites:



hysapupl.exe/out V8.1.1.91

db_sql/dbp_esk_sap_outbound_fm_post_error.hsc

Data field of the escalation	Value(s)	Meaning
SAP.FB	BAPI_REPMANCONF1_CREATE_MTS	Name of the function module in SAP
TYPE	'E' - for errors 'W' - for warnings 'A' - for interruptions	Subject to the business error
ID	Message ID	Subject to the business error
Number	Message number	Subject to the business error
Message	Message text	Subject to the business error

The function module BAPI_REPMANCONF1_CREATE_MTS has a return structure that is provided as export parameter. Consequently, only one error message may occur every time the function module is called.

9 SAP Order Sequencing

Overview

HYDRA menu	System administration → MES Link Enabling → SAP order sequencing
FEDRA menu	System administration → MES Link Enabling → SAP order sequencing
Transaction code	mleoss
Function authorization	mleoss.*

Purpose

Use the *SAP order sequencing* to control how the workplaces specified by SAP are transferred to the system. You can choose from the following options:

- The workplace transferred from SAP is interpreted as HYDRA group and the operation is planned in the pool of groups (backlog for machine group).
- The HYDRA group is selected for the workplace transferred from SAP and the operation is planned in the pool of groups (backlog for machine group).
- The HYDRA group is selected for the workplace transferred from SAP and the operation is directly planned for the machine.

This decision either affects the workplace or the entire system.

Integration

Diverse interfaces use these configurations to transfer orders from SAP.

Requirements

You have created workplaces and groups in the system.

Field descriptions

Key

Use this field to specify whether the entry applies to a specific order type or a machine.

In general, the configuration refers to a workplace.

Value

If you selected *order type* as the key, use this field to enter the order type the configuration applies for.

If you selected *machine* as the key:

- Enter a separate workplace in the *value* field if the configuration should apply for a specific workplace.
- Enter the value SYSTEM if you want the configuration to apply for the entire system.

Configurations referring to a specific machine take priority over the SYSTEM setting.



You can make a system entry for the majority of machines/workplaces/work centers and exceptions may be configured specifically.

Detailed planning

- **G**

"Transfer the SAP workplace as HYDRA group, plan operation in the pool of groups (backlog for machine group)."

- **M**

"Transfer the SAP workplace as HYDRA workplace, identify the HYDRA group, plan operation in the pool of groups (backlog for machine group)."

- **N**

"Transfer the SAP workplace as HYDRA workplace, identify the HYDRA group, and plan the operation for the workplace."

- **Rule 1 - Rule 9**

Use the rules 1 to 9 for customer-specific transfer logics, which will be implemented as part of the project.

10 Protecting fields of planned operations

Purpose

Use the configuration described in this document to prevent specific data fields of a (planned) operation from being overwritten when the operation is transferred once more via the ERP interface.



This function only affects ANR.MODIFY and/or ANR.UPDATE and operations.

Operations are only updated if the status of the order/operation generally allows it. The configuration described below does not apply if the status (see [order status assignment](#)) cannot be changed in general.

Requirements

You require the relevant function authorization to access [INI configuration](#) and [INI data configuration](#).

Procedure from service pack 12 onwards (b_anr.dll version 8.1.1.354)

Create a new entry in the [INI configuration](#):

Field name	Value
Name	BAPINOUPDATE
Description	Enter a description.

For this entry, create an entry including the following values in [INI data configuration](#):

Field name	Value
Section	ANR
Key	List the fields (HYDRA BAPI acronyms) that are not overwritten.
Value	The value includes a condition. Enter the condition, for example, as follows: ANR.ATYP=AG
Active	Yes

Use "@" to separate the single fields or conditions in the fields "key" or "value". The fields and conditions are processed one after the other.

You can define the values for "key" and "value" separately. The entries are processed one after the other. The conditions entered in the "value" field correspond to an AND operation.



As of service pack 12 only use the "@" character as separator if you create new entries or change existing ones. You do not have to change existing configurations (prior to service pack 12). In this case, the "|" character is still supported.



You can enter multiple entries for the function BAPINOUPDATE in the [INI data configuration](#), as you define the values for "key" and "value" separately.

Procedure up to service pack 11

Create a new entry in the [INI configuration](#):

Field name	Value
Name	BAPINOUPDATE
Description	Enter a description.

For this entry, create an entry including the following values in [INI data configuration](#):

Field name	Value
Section	ANR
Key	List the fields (HYDRA BAPI acronyms) that are not overwritten.
Value	Enter the condition that has to be met to make sure fields will not be overwritten. Enter BAPI acronyms including value.
Active	Yes

Use "|" to separate the single fields or conditions in the fields "key" or "value". The fields and conditions are processed one after the other.



Up to service pack 11 only use the "|" character as separator.

You can define the values for "key" and "value" separately. The entries are processed one after the other. The conditions entered in the "value" field correspond to an AND operation.



You can enter multiple entries for the function BAPINOUPDATE in the [INI data configuration](#), as you define the values for "key" and "value" separately.

If you cannot enter the pipe character ("|") using the GUI, you can still enter the values via the database:

- To do so, create a new entry as described above via the [INI configuration](#). Now use the following SQL statement to determine the internal DB counter for the header entry in the INI configuration:

```
select * from hyd_ini
```

- Determine the value of the "VERWEIS" column for the new entry.
- Create the required entries. Use the following SQL statement to assign the database table fields and application fields as described below:

```
insert into hyd_ini_data (ini_verweis, section, ident, value, bemerkung, aktiv)
values (<reference from previous SQL>, 'ANR', '<fields to be protected>', '<values>',
'<comment>', 'J')
```

Use the "|" (pipe) character to separate the acronyms of the fields you want to protect and the acronyms of the values.



Use a pipe character "|" to complete the list of the fields you want to protect and the list of values.

Database field	Values/content
INI_VERWEIS	The value of the VERWEIS column identified from the HYD_INI table via SQL.
SECTION	Section
IDENT	Key
VALUE	Value
BEMERKUNG (comment)	Comment
AKTIV	Active

List of frequently used acronyms

The following table lists the most frequently used acronyms and their meaning. Please contact MPDV Support if the list does not include the acronym you require.

Acronym	Meaning
ANR.MGRP	Machine group
ANR.MNR	Workplace/ machine
ANR.OPT:PLAN	Planning indicator: M Planned for workplace/machine G Planned for machine group
ANR.DATB	Start date planned (via HLS)
ANR.ZEIB	Start time planned (via HLS)
ANR.DATE	End date planned (via HLS)
ANR.ZEIE	End time planned (via HLS)

Example: protect the planned workplace

If the operation is planned on a workstation, you have to prevent the ERP interface from cancelling this planning. To do so, enter the below-mentioned data:

Field name	Value
Section	ANR
Key	ANR.MGRP@ANR.MNR@ANR.OPT:PLAN@
Value	ANR.ATYP=AG@ANR.OPT:PLAN=M@
Active	Yes

SLQ statement:

```
insert into hyd_ini_data (ini_verweis, section, ident, value, bemerkung,
aktiv) values (<reference from previous SQL>, 'ANR',
'ANR.MGRP@ANR.MNR@ANR.OPT:PLAN@', 'ANR.ATYP=AG@ANR.OPT:PLAN=M@', '<comment>',
'J')
```

Example: protect the start/end dates of a planned OP

If the operation is planned on a workstation and, as a result, its start time is specified, you have to prevent the ERP interface from cancelling this planning. To do so, enter the below-mentioned data:

Field name	Value
Section	ANR
Key	ANR.DATB@ANR.ZEIB@ANR.DATE@ANR.ZEIE@
Value	ANR.ATYP=AG@ANR.OPT:PLAN=M@
Active	Yes

SLQ statement:

```
insert into hyd_ini_data (ini_verweis, section, ident, value, bemerkung,
aktiv) values (<reference from previous SQL>, 'ANR',
'ANR.DATB@ANR.ZEIB@ANR.DATE@ANR.ZEIE@', 'ANR.ATYP=AG@ANR.OPT:PLAN=M@',
'<comment>', 'J')
```

11 MYERPRCK - Program Parameters

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Purpose

Use the upload program myerprck.exe/out to create confirmations/uploads to higher-level systems. In addition to the settings you make directly in the applications, you can also use program parameters to control confirmations/uploads.

Integration

The confirmation/upload is integrated with numerous components, for example:

- Shop floor data collection
- Tracking and tracing as well as material and production logistics
- Detailed scheduling

Available program parameters:

Parameters	Meaning/use	Relevant interfaces	Productive release
Program parameters to control processing:			
/MESTYP=XXXX	The parameter MESTYP defines the structure to be generated.	All	Yes
/GRP=XXXX	The grouping type specifies the criterion by which uploads should be grouped. Possible values: PLANT --> Groups by plant	Requires customizations	Requires customizations
/V=sssss	Since SAP R/3 PP does not support correction postings, HYDRA allows to retain confirmations/uploads for correction purposes in HYDRA for a specific period of time. Use the parameter /V=sssss (sssss =	EIS-ERP EIS-XPPS SAP-PPDC SAP-PPREM	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
	delay time in seconds) to activate the above described delay when the upload program is called. Examples: <pre>myerprck.exe/out /V=3600</pre> The system only uploads postings that are older than one hour.	SAP-PPPI SAP-PMCC3 SAP-PSCC4 SAP-COILV	
/BIS=DDMMYYHHMM /BIS=HHMM /TILLDATE=MM/DD/YYYY /TILLTIME=sec after midnight	Use the parameter /BIS=DDMMYYHHMM (date + time) when calling the upload program to enter the delay as a point in time. You can enter this point in time with date and time or you can just enter the time in the format "HHMM". In the latter case, the time refers to the current day. <pre>Myerprck.exe /BIS=2505110600</pre> This parameter uploads postings that were recorded until 06:00 a.m. on 25 May 2011. <pre>Myerprck.exe /BIS=0600</pre> This parameter uploads postings that were recorded until 06:00 a.m. of the current day.	EIS-ERP EIS-XPPS SAP-PPDC SAP-PPREM SAP-PPPI SAP-PMCC3 SAP-PSCC4 SAP-COILV	Yes
/TZ=+/-sssss	Use the parameter /TZ=+/-sssss to adapt uploads to different time zones. The parameter adjusts the time specifications entered in the fields EXEC__START_TIME, EXEC_FIN_TIME and LOGTIME of the upload structure of the SAP-PPDC interface according to its specifications.	SAP-PPDC	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
/KST=XXX	Use this parameter to restrict the data to be uploaded. In this case, the system only uploads data of a specified cost center. Use the parameter /KST=XXX (XXX = cost center, a max. of 8 characters) when calling the upload program myerprck.exe/out to enable the above-described restriction. Then the system only uploads data records that were posted to machines of the specified cost center. The system checks the cost center of the machine/workplace that is entered as the posting workplace/machine in the posting record. The system only checks the cost center of the workplace/machine. You can specify the parameter several times per call. <u>Example:</u> <pre>Myerprck.exe /KST=BDE100 /KST=BDE200</pre> The system only uploads records that were posted onto machines of the cost center BDE100/BDE200.	EIS-ERP ESI-XPPS SAP-PPDC SAP-PPREM SAP-PPPI SAP-PMCC3 SAP-PSCC4 SAP-COILV	Yes
/CLEAR_RES	Use the parameter "/CLEAR_RES" to assign an "X" to the field CLEAR_RES of the upload structure when it comes to a final confirmation/upload (record type L40). Consequently, SAP will clear open reservations for the respective order.	SAP-PPDC	Yes
/NEG_MENGE	By default, quantities (L20/L40) cannot	SAP-PPDC	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
	be uploaded to SAP PP using partial confirmations/uploads via the SAP-PPPDCC interface if data is collected at the same time via the total quantity counter of MDE machines, since SAP is not able to process negative quantities. This type of collection can result in negative quantity postings for yield when OPs are finished. This restriction does no longer apply, if it is possible to process such negative postings (e.g. by using the SAP standard BAPI or customizations). In this case, you can use the program parameter /NEG_MENGE to enable the upload of these quantities.		
/LA_MNR	The SAP_PMCC3 interface requires the activity type to be uploaded to SAP PM. The activity type can be identified via the machine/workplace where the posting was performed. Use this program parameter to enable identification of the activity type. Then the system uses the machine to identify the activity type from the activity types kept in HYDRA.	SAP-PMCC3	Yes
/IDENT_PRAEFIX=	In the upload structure of the SAP-PPPDCC interface, the field EX_IDENT uniquely identifies uploads from subsystems. HYDRA populates the field. You can add a prefix to the EX_IDENT field to differentiate between uploads	SAP-PPPDCC SAP-PPPDCC	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
	from various HYDRA subsystems connected to one SAP instance. <u>Example:</u> <pre>Myerprck.exe /IDENT_PRAEFIX=ABC</pre> The prefix may only include hexadecimal characters: A –H und 0 – 9.		
/ABZEICH=XX	While customizing the order type, you can specify that only signed data records are uploaded. Use the parameter /ABZEICH=XX to specify a period of time in days after that you can upload even unsigned data records.	EIS-ERP EIS-XPPS SAP-PPDC SAP-PPREM SAP-PPPI SAP-PMCC3 SAP-PSCC4 SAP-COILV	Yes
/TRANSFER=	Use the parameter "/TRANSFER=" to only upload records whose specifications were transferred from a specific system. The transfer indicator is set during HYDRA inbound processing and may vary from interface to interface.	EIS-ERP ESI-XPPS SAP-PPDC SAP-PPREM SAP-PPPI SAP-PMCC3 SAP-PSCC4 SAP-COILV	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
/NOTTRANSFER=XXX	Use the parameter "/NOTTRANSFER=" to only upload records whose specifications were NOT transferred from a specific system. The transfer indicator is set during HYDRA inbound processing and may vary from interface to interface.	EIS-ERP ESI-XPPS SAP-PPDC SAP-PPREM SAP-PPPI SAP-PMCC3 SAP-PSCC4 SAP-COILV	Yes
/SEK	The EIS-ERP interface uploads the times of resource performance accounts in hours. In particular with very short lead times this may effect that logon times are cut off by a conversion into hours. Use this program parameter to upload times in seconds.	EIS-ERP ESI-XPPS	Yes
/RMTYP=	When customizing the order type, you can assign an upload type to the order type. Use this program parameter to only upload data records of this upload type. You can specify the parameter several times per call.	EIS-ERP ESI-XPPS SAP-PPDC SAP-PPREM SAP-PPPI SAP-PMCC3 SAP-PSCC4 SAP-COILV	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
/KAT=	When customizing the order type, you can connect the order type with a category. Use the program parameter /KAT= to only upload data records of this category. You can specify the parameter several times per call.	EIS-ERP ESI-XPPS SAP-PPDC SAP-PPREM SAP-PPPI SAP-PMCC3 SAP-PSCC4 SAP-COILV	Yes
/SART=	The system only uploads ADE log postings of the specified record type. Therefore, you can use different program parameters per call and record type for uploading. Requirement: You have to activate the corresponding uploads when customizing the order type. You can specify the parameter several times per call. <u>Example:</u> <pre>Myerprck.exe /SART=A /SART=E</pre> ⇒ The system only uploads A and E records.	EIS-ERP ESI-XPPS SAP-PPDC SAP-PPREM SAP-PPPI SAP-PMCC3 SAP-PSCC4 SAP-COILV	Yes
/NOLOCK	When starting the upload program, the system checks if there are any lock entries for the database table ADE_PROTOKOLL. If this is the case,	All	Requires customizations

Parameters	Meaning/use	Relevant interfaces	Productive release
	the upload is not carried out. You can use this program parameter to prevent this check. Set this parameter, in particular, if the upload is not based on the table ade_protokoll.		
/EINH_CC34	The interfaces SAP-PMCC3 and SAP-PSCC4 transfer the uploaded activity quantity in seconds (SEC) to SAP. Use the parameter "/EINH_CC34" to upload the data in other units. The following units are supported: Hours: H, HUR, STD Minutes: MIN Seconds: SEC <u>Example:</u> Myerprck.exe /EINH_CC34=HUR The system uploads the recorded times in the unit "HUR" (hours).	SAP-PMCC3 SAP-PSCC4	Yes
/SDAT_STORNO	The SAP-PPPDCC interface transfers the change date along with the correction records. Use this program parameter to upload the initially collected shift date instead.	SAP-PPPDCC	Yes
/NORFC_STORNO	The SAP-PPPDCC interface transfers the cancellation records via sRFC. Use the program parameter to transfer the data in the IDoc format to SAP. To do	SAP-PPPDCC	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
	so, inbound processing must be implemented in SAP. The system uploads the cancellation records via the standard PP-PDC segment (with record type K20/K40) as if the PP-PDCC license was not available.		
/PI	If you use the SAP Process Integration (previously: Exchange Infrastructure) to communicate with SAP, the version of the transferred segment is checked more strictly. Use the program parameter to transfer segment names with the version number (i.e. the trailing zeros of the segment name).	SAP-PPDC SAP-PMCC3 SAP-PSCC4	Yes
/INDEX_TMP_TABLE	Use this parameter to accelerate uploads if ORACLE is used as database system and large amounts of data are affected. To do so, use an index for a temporary table where all data to be uploaded is transferred in a first step.	All	Requires customizations
/UE_PARAMS=	Program parameter for the stand-alone user exit processing (DD format).	Various	Yes
/NOSTORNO	Use this program parameter to prevent cancellation records from being uploaded. Therefore, you can use different program parameters per call and record type for uploading. Requirement: You have to activate the	All	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
	corresponding uploads when customizing the order type.		
/RECALC_NEG_YIELD	Use this parameter to offset negative yield with already posted positive uploads.	SAP-PPPDCC	Requires customizations
Program parameters to use the SIGUSR communication:			
/LOGGING	Use this program parameter to activate communication from the database table HYD_LOGGING. To do so, a customization might be required.	INDIVIDUAL CASE	Yes
/WAIT_SIGUSR1=XX	The program parameter specifies the time in seconds that has to pass before the upload is performed via the SIGUSR communication even without trigger.	INDIVIDUAL CASE	Yes
/PEEK_SIGUSR1=XX	Use this parameter to delay execution of an action triggered by the SIGUSR communication. The delay time is entered in seconds for this parameter. The program interprets this time as follows: If within the next second after the initial trigger there is another trigger, then wait for not more than <specified value> seconds. If in a specific case, triggers would indeed arrive every second then the	INDIVIDUAL CASE	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
	WAIT_SIGUSR time (e.g. 120 seconds) would apply; i.e. the system would in fact perform the upload after 2 minutes.		
/SEND_SIGUSR1=	This program parameter defines which other process/ program must be triggered after processing by the SIGUSR communication. Specify the process/program WITHOUT file extension.	INDIVIDUAL CASE	Yes
/COUNT_SIGUSR1=XX	Uploading in signal mode can hardly be subjected to tracing. This is due to the fact that the program in those cases is started once via the scheduler but won't shut off. Any redirection of the program call with -d to a log file will then necessarily lead to very large log files, which will negatively affect the performance. Use the new program parameter /COUNT_SIGUSR1=XX to specify after how many calls the program will automatically shut down. A call in these instances is both, a call via SIGUSR communication and the cyclical program execution which is controlled via the parameter /WAIT_SIGUSR1. Then the scheduler restarts the program. But this will lead to a time period "t" during which SIGUSR calls will not be processed. It is, however, assumed that this will not lead to data losses since the data to be uploaded are already saved to	INDIVIDUAL CASE	Yes

Parameters	Meaning/use	Relevant interfaces	Productive release
	the DB. Benefits: If the program is started via a script (*.scr) from the scheduler, you can store there the routine to generate a date/ time stamp file name for the log file to be created. This allows to restrict the log file size.		
Program parameters for debugging/ tracing/ testing/ logging purposes:			
/ONLYERR	This program parameter specifies that system log entries are only created if an error occurred during uploading. This reduces the entries in the system log.	All	Yes
/SIM	The system does not upload/confirm data during simulations (the uploaded/confirmed indicator is set to "True").	All	No
/SIMULATION	The system does not upload/confirm data to SAP during simulation (confirmed/uploaded indicator will not be changed).	All	No